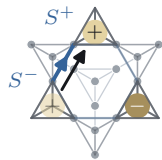


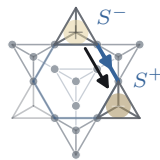
i.a) create


 $S_i^+ S_j^-$: pair splits

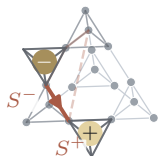
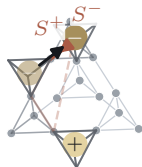
i.b) move


 $S_i^+ S_j^-$: spinon walks

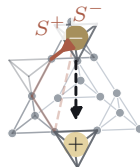
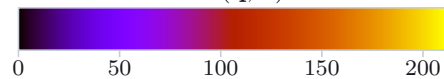
i.c) annihilate


 $+$ meets $-$
 $\mathbf{q}_\gamma = (0, 0, 0)$

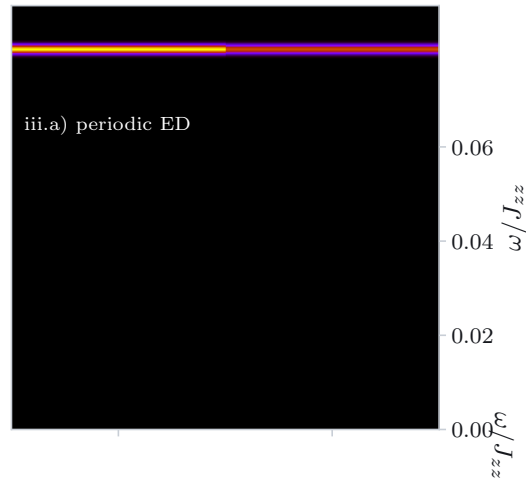
i.d) create


 $S_i^+ S_j^-$: pair splits
i.e) apply $S_i^+ S_j^-$ 
 carried out of the cell
 the boundary returns it
 $\mathbf{q}_\gamma = (0, 0, -1)$

i.f) identify


 $S^{zz}(\mathbf{q}, \omega)$


iii.a) periodic ED



iii.b) winding-free

